Customer Analytics: Using Machine Learning to Predict the Customer Life Time Value (CLTV) Based on Purchase History and Ratings

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***Abstract*— Customer Life Time Value (CLTV) is a vital customer analytics metric that is intended to inform companies on the right investment in customer acquisition, retention, and engagement. Conventional CLTV models are heavily dependent on purchase frequency and money-value transaction information, but not customer satisfaction measurements like ratings. This paper proposes a general framework for incorporating ratings into machine learning models for CLTV prediction. Taking advantage of the merits of ensemble learning methods, the paper uses both transactional and satisfaction data to uncover the complex customer behavior dynamics. Our experimental finding indicates the better predictive performance of satisfaction-based models, providing companies with a sound basis to enhance customer analytics activities.**

Keywords—Customer Lifetime Value (CLTV), Customer Analytics, Machine Learning, Customer Ratings

I. Introduction

Customer Life Time Value (CLTV) determines the lifetime revenue a customer will bring under a period of interaction with an enterprise. Correct estimation of Customer Lifetime Value (CLTV) is important for companies to achieve profitability and customer satisfaction. Recency, Frequency, and Monetary (RFM) analysis using transactional data is a conventional method and fails to capture the engagement and satisfaction dimensions that make up customer value. Presence of customer feedback metrics like ratings provides a chance to look deeper into customer satisfaction and intent. Ratings indicate customer engagement and loyalty and are actionable measures of customer behavior as well. Although these metrics are less leveraged in CLTV estimation, their absence may lead to the false assumption of high-value customers who always provide positive ratings. This research suggests a prediction model of CLTV based on ratings that leverages transactional and satisfaction data using advanced machine learning algorithms. Overcoming the limitations of existing CLTV models, this research tries to offer a comprehensive and scalable method to achieve better estimation of CLTV, following the evolving requirements of companies and promising the leverage of satisfaction metrics in customer analysis.

II. Literature Review

Customer Life Time Value (CLTV) prediction has evolved from simple statistical prediction to advanced machine learning (ML) models, and more recently, to leveraging of customer satisfaction measures, i.e., ratings. This part highlights the recent past of CLTV research development, with an emphasis on key breakthroughs as well as limitations that necessitate the development of rating-based models.

A. Traditional CLTV Models

Traditional CLTV models were established on the basis of early methodologies for estimating customer value from transactional data. The first such framework was Recency, Frequency, and Monetary (RFM) analysis, which divided customers by purchase history. Recency examined the date of the last purchase, recent history implying future activity; Frequency examined the number of purchases in a time interval, repeat buyers being high-value; and Monetary Value examined total spending, more spending implying greater profitability. This was easy to compute, extremely interpretable, and computationally efficient. It had the drawback of not examining non-transactional behavior such as satisfaction or intent and was therefore less suitable in dynamic circumstances.

Probabilistic models such as Pareto/NBD augmented standard RFM analysis by making statistical assumptions of customer retention and churn [2]. These models generated estimates of the likelihood of customers staying active and predicted total spending before churn and thus offered a probabilistic estimate of customer lifetime value. These models, though augmenting, were limited by fixed assumptions of customer behavior, and thus were lagging in accommodating changes in satisfaction or preferences in real time. Similarly, Buy-Till-You-Die (BTYD) models, such as the BG/NBD model, mixed probabilities of churn with past purchase activity to predict future transactions. Though these models were an improvement on simple RFM in incorporating temporal retention probabilities, they did not incorporate qualitative metrics like customer satisfaction, which would have the potential to impact predictions greatly.

B. Machine Learning in CLTV Prediction

Machine learning has transformed CLTV prediction by allowing the estimation of complex, nonlinear relationships among features. Random Forests, being ensemble learning algorithms, combine predictions from an ensemble of decision trees, dealing with high-dimensional datasets with ease and computing feature importance. They are thus useful in determining what variables, e.g., ratings or purchase frequency, affect predictions. They may be less useful for sequential data, e.g., temporal dynamics of customer ratings. Gradient Boosting Machines (GBM) such as XGBoost and

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LightGBM algorithms construct decision trees iteratively to reduce prediction errors. Their high predictive power is due to iterative optimization of residuals, and they are well suited to combining heterogeneous features, e.g., transactional and behavioral features. GBMs, however, overfit in the absence of regularization and can be computationally intensive on large datasets. Neural Networks, conversely, represent complex patterns by learning representations of latent features. They are well suited to combining transactional features with demographic and engagement data, capturing nonlinear dependencies such as the interaction between ratings and purchase frequency. Neural Networks, however, need large datasets and fine tuning and are typically not interpretable, and extracting insights from them can be difficult.

C. *Deep Learning and Sequential Modeling*

Deep learning models, especially sequential data-specific models, have greatly enhanced the prediction of Customer Lifetime Value (CLTV). Long Short-Term Memory (LSTM) is best suited to model long-range dependencies and thus is particularly well-suited to handling sequential data, e.g., purchase history or trend in ratings [1]. Such networks are highly effective at identifying the temporal patterns in customer satisfaction and understanding how the fluctuation in ratings affects future value. They are thus particularly useful in industries that are subject to seasonal buying behavior or customer loyalty-driven trends. They are computationally expensive and susceptible to overfitting unless combined with strong regularization methods. Transformers, however, with their use of self-attention mechanisms, have proven to be useful sequencing and multimodal fusion tools [5]. They focus on the relative importance of different features, e.g., the importance of satisfaction metrics to CLTV valuation, and are able to handle complex and large data sets such as customer reviews.

D. *The Role of Customer Ratings in CLTV*

While transactional data continues to be the centerpiece of CLTV estimation, ratings add richness by placing a value on satisfaction and intent. Ratings offer a point-in-time measure of the sentiment customers hold about their experiences, connecting satisfaction to future expenditure and loyalty. Customer satisfaction is a stable precursor to financial performance, for example, and a must in predictive models. Trends over time in ratings show shifts in customer attitudes and are leading indicators of churn or growing loyalty [3]. Models that are capable of measuring these trends are more capable of learning about changing customer behavior. In addition, explainable AI methods, like SHAP values, have been applied to measure the impact of satisfaction metrics, demonstrating their underutilization in conventional CLTV models [4].

E. *Addressing Research Gaps*

Even with such advances, there remain fundamental gaps in CLTV prediction. Ratings, while promising, get relegated to secondary features instead of being used as first-order predictors, and their full power is not harnessed. State-of-the-art models such as deep learning are susceptible to being black boxes, and actionable insights from satisfaction-based predictions become problematic. These gaps are addressed using a unified framework that places ratings at the center of CLTV prediction but remains scalable and explainable.

F. *Comparison of Commonly Used Algorithms*

Several ML algorithms are commonly used for CLTV prediction, each with its strengths and weaknesses:

TABLE 1: ALGORITHM COMPARISON

Algorithm	Strengths	Weaknesses
Regression Analysis	Simple to implement, interpretable results.	May not capture complex non-linear relationships.
Gradient Boosting Machines	High predictive accuracy, handles complex relationships.	Can be computationally expensive, prone to overfitting.
Survival Analysis	Useful for understanding customer retention and predicting future revenue streams.	Requires specific data format and assumptions.
Decision Trees	Easy to interpret, handles both categorical and numerical data.	Can be unstable and prone to overfitting.
Neural Networks	High predictive power for complex data.	Requires large datasets, computationally expensive
Markov Chains	Captures sequential patterns and transitions in customer behavior.	Can be limited by the assumption of state independence.

III. METHODOLOGY

The predictive approach for Customer Life Time Value employs feature engineering, complex machine learning algorithms, and rigorous validation processes. Transactional data and rating data are both considered in this process so that all transactions receive a representation in these ratings and therefore receive a comprehensive view of customer activity. The subsequent sections elaborate on the different components involved:

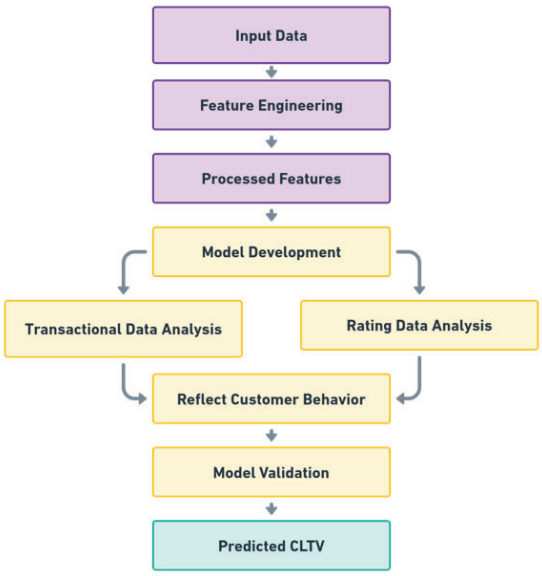

Fig.1 Model Framework

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A. Feature Engineering

Feature engineering is vital for transforming raw data into useful predictors. Therefore, it will be important to integrate ratings alongside traditional transactional variables. The dataset analysis ensures that features selected give meaningful insights while being free from redundancy and collinearity.

Ratings Metrics

Ratings measure customer satisfaction and provide a more comprehensive insight of their preferences and brand affinity. These indicators were preprocessed and aggregated at the individual and temporal levels to show long-term patterns. The key rating metrics are **Average Customer Ratings and Rating Trends Over Time, Transactional Metrics** are foundational to understanding purchase behaviors and predicting revenue-related outcomes:

TABLE 2: PARAMETERS

Parameters	Description
Purchase Frequency	Number of transactions over a defined period of time
Average Order Value (AOV):	Amount a Customer spends by calculating the average monetary value of their orders.
Days Since Last Purchase:	Shows how recently purchases were made
Average Customer Ratings	Divided into "High," "Medium," and "Low" categories using quantile-based segmentation.

Average Customer Ratings: **High** means the Top 25%, **Medium** means the Middle 50% and **Low** means the Bottom 25% of ratings.

Parameter Evaluation and Reduction

In feature engineering, all the parameters in the dataset are taken into account to see if they are independent and relevant. After correlation analysis and domain knowledge, parameters that were not actually independent predictors were dropped. For example, spend per Item, this variable was initially taken into account but subsequently dropped because it is highly correlated with total spend and also with purchase frequency. Including it in the model would have potentially resulted in multicollinearity, which would have adversely affected the performance of the model. This screening process enabled us to keep only the most significant predictors, so that the ML model could concentrate on features that make independent and unique contributions towards the prediction of CLTV.

B. Model Development

In order to encapsulate the complex interdependencies that exist between ratings data and transactional data, ensemble learning models like the Random Forest Regressor and Gradient Boosting Regressor were employed. Both these models were chosen because of their proven ability to handle nonlinear relationships in high-dimensional data effectively.

Model Selection and Training

1. **Random Forest Regressor**: A non-parametric ensemble method, Random Forest constructs many decision trees during training and outputs the average prediction of the trees, thereby reducing variation and preventing overfitting. Particularly helpful with datasets that have many features, as it uses feature importance to rank predictors.
2. **Gradient Boosting Regressor**: An iterative ensemble technique in which weak learners (decision trees) are sequentially learned to reduce prediction errors. Prevents overfitting using methods such as shrinkage (learning rate) and regularization (minimum samples per split). Renowned for its strong predictive capabilities and aptitude for detecting subtle patterns in data. Hyperparameter Tuning The two models were hyperparameter tuned with GridSearchCV to determine the optimal combination of hyperparameters.

Random Forest Parameters:
The hyperparameter tuning optimized key parameters: n_estimators (100, 200), max_depth (10, 20), min_samples_split (2, 5), and min_samples_leaf (1, 2), ensuring a balance between model complexity and performance.

Gradient Boosting Parameters

Gradient Boosting was tuned with the following parameters: n_estimators (100, 200), max_depth (3, 4), learning_rate (0.01, 0.1), and min_samples_split (2, 5) to optimize performance.

Using 5-fold cross-validation, both models were trained on scaled training data and were evaluated on the test set for performance assessment.

Mathematical Representation of the Models

Random Forest Regressor: It is an ensemble method that constructs many decision trees and combines their predictions. The final prediction, ŷ, is derived as the average of predictions from M individual trees.

$$y^{\wedge} = M1\ i=1\sum M\ hi\ (x) \qquad (1)$$

Where **hi (x)** represents the prediction from the **i-th** decision tree, **M** denotes the total number of trees, and **x** is the input feature vector.

Each decision tree is then trained on a randomly sampled subset of the data using bootstrap sampling, and a random subset of features is considered at each split. This approach reduces correlation between trees, enhancing generalization and preventing overfitting.

In **Gradient Boosting Regressor**, models are built sequentially, with each iteration refining the previous model's errors. The prediction at iteration **t**, denoted as $\mathbf{h_t(x)}$, is updated as:

$$ht\ (x)=ht-1\ (x)+\eta\cdot gt\ (x) \qquad (2)$$

Where **ht (x)** is the Prediction at iteration t, **η** is the Learning rate, controlling the step size, **gt (x)** is the Gradient of the loss function L regarding predictions ht−1 (x), given by:

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$$g_t(x) = -\partial h_{t-1}(x) \partial L(y, h_{t-1}(x) \quad (3)$$

The final prediction is the sum of all individual learners:

$$\hat{y} = \sum_{t=1}^{T} \eta \cdot g_t(x) \quad (4)$$

Where T is the total number of iterations (trees).

IV. Results

Model Performance

The models were assessed using key performance metrics such as R^2 Score, Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). The evaluation results are summarized below.

TABLE 3: Model Performance

Metric	Random Forest	Gradient Boosting
R^2 Score	0.9995	0.9997
RMSE	8.28	5.98
MAE	2.45	1.39
MSE	68.58	35.70
MAPE (%)	0.19	0.11

Insights

- **Gradient Boosting Regressor** demonstrated superior performance, with a lower RMSE (5.98) and MAPE (0.11%) compared to Random Forest. This indicates better predictive accuracy and lower average errors.
- **Random Forest Regressor** provided strong results, especially with its explainability through feature importance analysis, but lagged slightly in precision compared to Gradient Boosting.

Future Directions

While ensemble methods provided robust results, future work can explore:

- **Deep Learning Architectures**: Multimodal models combining feed-forward neural networks for structured transactional data with convolutional neural networks (CNNs) for unstructured engagement data.
- **Attention Mechanisms**: These can highlight the most influential predictors, enhancing both performance and interpretability.

Visualization and Interpretability

Importance of predictors was established based on feature importance scores produced by the two models. Visual plots such as scatter plots, residual plots, and histograms showing error distributions were informative on the performance and error nature of the models.

This methodology focuses on merging interaction and transactional data in a well-designed machine learning system. Merging the data types increases the predictive power of the method, giving more useful data on the character of customer behavior. The heatmap visualizes the correlation coefficients between different dataset features, coefficients range from -1 to 1. A value of 1 denotes a 100% positive correlation, -1 represents a 100% negative correlation, and 0 signifies no correlation between the features.

Observations :

1. **Strong Positive Correlations**:
   - **Average Rating and Total Spend**: The correlation coefficient is around ~0.94, showing an extremely high positive relationship that suggests customers with a higher rating spend more.
   - **Membership Type and Gender**: A significant correlation (~0.81) between these features may reflect underlying relationships tied to customer demographics and membership choices.
2. **Moderate to Weak Correlations**:
   - **Age and Total Spend**: There is a moderate negative correlation (~-0.68), indicating that younger customers may spend more than older customers.
   - **Days Since Last Purchase and Satisfaction Level**: A correlation (~0.72) suggests that satisfaction levels might impact the frequency of purchases.
3. **Low Correlations**:
   - **Satisfaction Level and Total Spend**: A negligible negative correlation (~-0.045) indicates little to no direct relationship between these features.
4. **Negative Correlations**:
   - **Age and Average Rating**: A notable negative correlation (~-0.72) indicates that older customers might rate their experiences lower on average.

Significance for Feature Engineering: This heatmap is a critical tool for identifying redundant or dependent features.

- **Membership Type and Gender**: Given their high correlation (~0.81), including both might introduce multicollinearity into the model.
- **Average Rating and Total Spend**: While highly correlated, both metrics may provide unique value in the context of transactional and behavioral analysis.

By examining these relationships, features with poor prediction power or high interdependence with other features can be removed or reformed to improve model performance and minimize redundancy. This process guarantees the inclusion of independent, relevant predictors in CLTV prediction. The scatter plots depict the relationship between actual values (x-axis) and predicted values (y-axis) for the Random Forest Regressor (Fig. 3a) and Gradient Boosting Regressor (Fig. 3b). The red line represents the scenario where predictions align perfectly with actual values.

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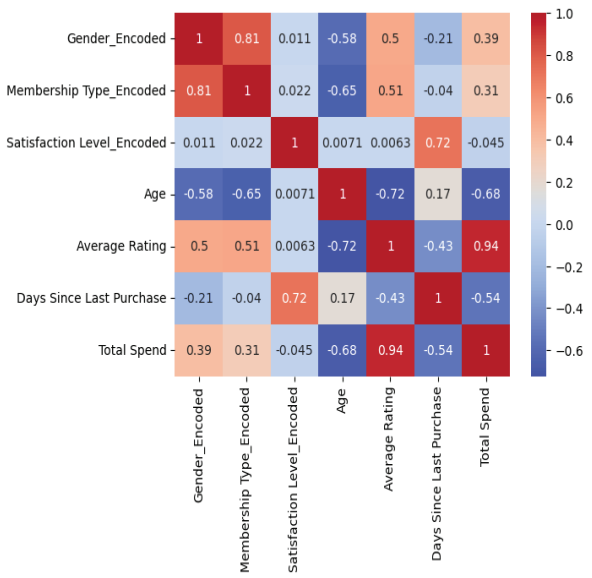

Fig 2. Heatmap

Observations :

1. **Alignment with the Ideal Line**: Both models exhibit an almost perfect alignment with the ideal red dashed line, indicating high predictive accuracy. The Gradient Boosting Regressor exhibits a tighter clustering around the ideal line, indicating a better alignment between predicted and actual values. This suggests that it outperforms the Random Forest Regressor in terms of prediction accuracy.

2. **Minimal Deviation**: The points in the figures show minimal deviation from the red dashed line, indicating low residual errors. This suggests that both models effectively capture the relationships between the features and the target variable, demonstrating strong predictive performance.

3. **Outliers**: Neither graph has any outliers, meaning that both models demonstrate excellent generalization throughout the dataset without succumbing to overfitting or underfitting.

Significance: The plots confirm the high R^2 values achieved by both models, with the Random Forest Regressor attaining 0.9995 and the Gradient Boosting Regressor achieving 0.9997. The closer fit seen in the Gradient Boosting Regressor validates that its sequential learning process effectively reduces residual errors, thus improving precision. This claim is further supported by its lower RMSE and MAPE values, which show reduced prediction errors. The congruence of predicted values and actual data, with little deviation and no extreme outliers, validates the precision and reliability of both models. All these collectively establish that the models provide highly accurate and reliable CLTV predictions and are well-suited especially for real-world customer analytics use cases. The residual plots given below show the actual-predicted value differences (residuals) for the Random Forest Regressor (left) and the Gradient Boosting Regressor (right).

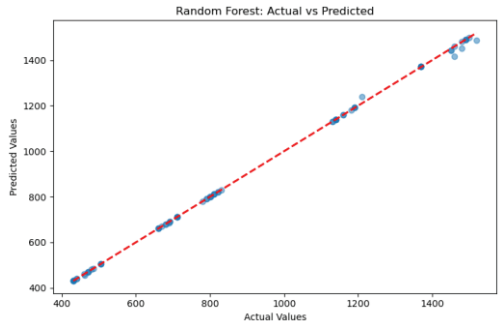

Fig 3a. Random Forest: Actual vs Predicted

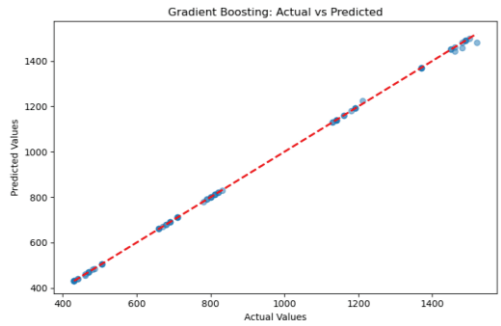

Fig 3b. Gradient Boosting: Actual vs Predicted

Observations :

1. **Centered Residuals**: The residuals for both models are centered around the red dashed line (y=0), which means no bias in the predictions.

2. **Random Forest**: Shows a bit higher residual, particularly for larger predicted values, reflecting occasional over or under-predictions.

3. **Gradient Boosting**: Residuals are tighter and more closely packed around zero, reflecting precision in predictions compared to Random Forest.

V. Conclusion

This study shows the necessity of considering customer subjective perceptions to enhance the precision and relevance of Customer Lifetime Value (CLTV) prediction models. Both the models show strong performance in CLTV prediction. However, the Gradient Boosting model outperforms the Random Forest model, yielding lower residuals, Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE) values.

Residual Analysis: Residual plots for both the models show stark contrast in performance.

- **Gradient Boosting:** The residuals are close to zero, depicting that the predictions are highly accurate and centered around the actual values and they have low bias.
- **Random Forest:** While effective, it exhibits slightly larger residuals, particularly for higher predicted values, suggesting occasional over- or under-predictions.

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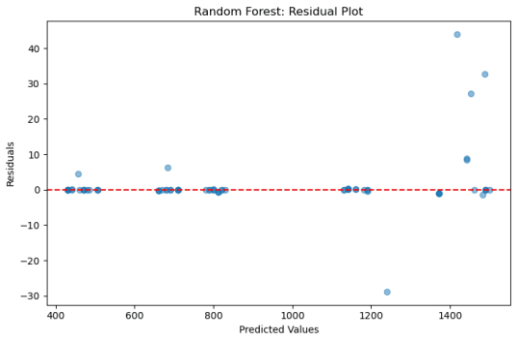

Fig 4a. Random Forest: Residual Plot

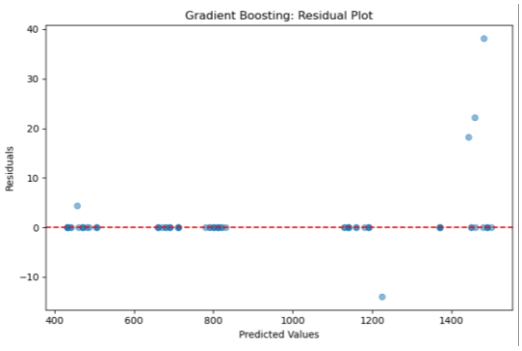

Fig 4b. Gradient Boosting: Residual Plot

Spending Behavior by Rating Segments: The analysis of customer spending by rating segments shows very strong correlation between satisfaction and spending:

1. **Top 1/3 Average Ratings:** At the top 1/3 of the best-rated customers, they have the highest average spend ($1309.17), which is great compared to the overall average of innovation $845.38. This confirms the positive relationship between satisfaction and income.
2. **Middle 1/3 Average Ratings**: Somewhat satisfied consumers are lower spenders ($752.12) compared to very satisfied customers, illustrating the need to add value to experiences to drive participation.
3. **Bottom 1/3 Average Ratings:** Disparaged customers have the least (average spend: $473.39) highlighting that failing to ensure customer satisfaction can be extremely costly.

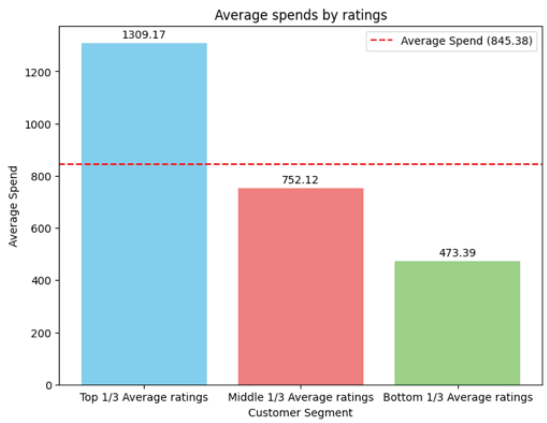

Fig 5. Average Spends by ratings

Key Insights and Implications: The results clearly demonstrate that with greater satisfaction comes greater spend, and that customer rating is a significant predictor variable in CLTV models. Organizations should concentrate their efforts on driving customers upwards from the Middle and Bottom rating categories to achieve the Top category by geo-targeted offers, and service improvement. (3) Adding customer ratings as a weighted feature in CLTV models allows you to make them more predictable and closer to real customer behavior.

Future Directions

- **Unstructured Data Integration:** Using text reviews and social media sentiment to augment satisfaction metrics.
- **Advanced Architectures:** Attention mechanisms and deep learning models for feature interpretability and scalability.
- **Cross-Industry Validation:** Using the framework in another sector to check its generalization ability and adaptability.

This study takes a practical approach, highlighting customer satisfaction as one of the most significant predictors in CLTV models, allowing firms to identify areas to invest in, in order to maximize retention and revenue. The ratings are an opportunity to play a game-changing role with better predictive abilities and customer-centric initiatives.

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